

round-the-clock by the  $24 \times 7$  local operation department and regional operation centre.

While the IT department started exploring several protocols like telnet, wget, ftp, web database, etc, I suggested uploading it to a cloud wherefrom it could be accessed by all. But doing it on Arduino Mega is a challenge, though not impossible.

## ESP32 LoRa router

Its rather easy to achieve this feat through ESP32. The UART LoRa radio is first to be connected to the receiver/transmitter of the ESP32. While softwareserial works on Arduino, only hardwareserial works on ESP32. Therefore, the LoRa radio needs to be connected on the hardware serial port first. For setting hardwareserial on ESP32, the following command is used:

```
Serial1.begin(9600, SERIAL_8N1, RXD1, TXD1); //
Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2); //
// For Rx = 16,15,13,34
// For Tx = 17,12,4,35
```

You can set your own pins for the transmitter and receiver. The above-mentioned pins worked splendidly. For connecting two LoRa radio units two UART sets can be used. As per data book, ESP32 can have three UARTs at a time and any GPIO pin can be set for serial communications.

Once the LoRa radios start communicating with the ESP32 on the UARTs, the next step is of setting up the Wi-Fi network for connecting with the internet. This crucial part of networking can be done through the command:

```
#include <WiFi.h>
#include <WiFiUdp.h>
#include <HTTPClient.h>
#include <esp_wifi.h>
const char *ssid = "beral";
const char *password = "*****";
const char *ssid1 = "bera2";
const char *password1 = "*****";
```

I have two networks to use and,

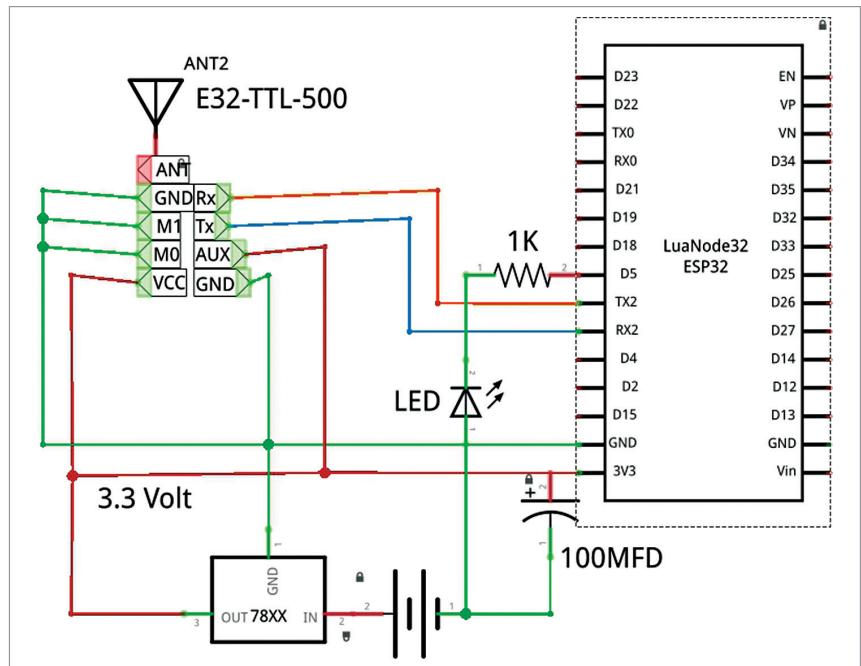


Fig. 3: Circuit diagram of ESP32 LoRa gateway

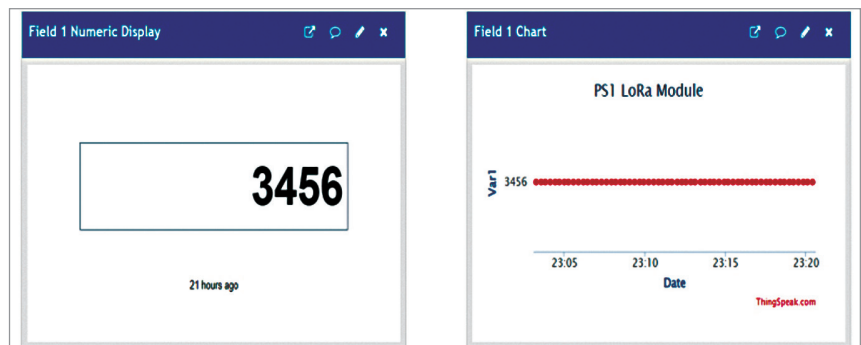


Fig. 4: Screenshot of the data

depending on the connectivity, the ESP32 is programmed in such a way that if it does not get the connection, it looks up its EEPROM memory and restarts to connect to an alternate network. You can provide any number of SSID and passwords and, in case it fails to connect, the ESP32 will try all of them one after the other. Of course, small changes for additional network selection criteria would need to be adopted in the setup area.

When the ESP32 starts getting data at the SS radio network and is connected to the internet using Wi-Fi or BLE (Bluetooth Low

Energy), if we can transfer the data from one network to the other, the ESP32 will achieve the status of a router. At this point we can select our own server or cloud, or we can find a third-party server or cloud. Since our company's cloud is in the making, we decided to upload eight sets of very critical data to a public cloud for distribution and visibility for everyone. The data is available at following site at present: <https://thingspeak.com/channels/279012>

## Circuit and working

The pin connections of the ESP32 LoRa gate are shown in Fig. 3. The